Project Phase 2: ER Model

Aegis Intelligence Database (AID)

Team Name: Big Three

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1 Changes from Phase 1

During the refinement of our ER model, we made several strategic modifications to improve data integrity and accurately represent the relationships within the Aegis Intelligence Database:

1.1 Structural Changes

- **Bounty Attribute Migration:** Moved the `Bounty` attribute from the `PIRATE` entity to the `BOUNTY_RECORD` weak entity. This change reflects the reality that a pirate's bounty changes over time, and each bounty update represents a new record with its own issue date and amount. This prevents data redundancy and maintains historical bounty information.
- **Role Attribute in MEMBERSHIP:** Added a `Role` attribute to the `MEMBERSHIP` relationship to specify a person's role within a crew (e.g., "Captain", "Doctor", "Navigator", "Sniper"). This provides crucial contextual information about crew structure and individual responsibilities.
- **Identifying Relationship HAS:** Added the identifying relationship `HAS` to properly connect the `PERSON` entity with the weak entity `BOUNTY_RECORD`. This formalizes the dependency relationship and ensures that bounty records cannot exist without their associated person.
- **Home Island Foreign Key:** Changed the `Home_Island_ID` from being specific to `CIVILIAN` to being an attribute of the general `PERSON` entity. This modification recognizes that all individuals, regardless of their classification (Pirate, Marine, or Civilian), have a place of origin that is strategically valuable intelligence.

2 ER Diagram Notation

Due to the limitations of our diagramming tool (Mermaid), we were unable to strictly follow all standard ER notation conventions, but we tried to keep as much standard notation as possible. Below is our notation scheme used in the diagram:

2.1 Entity Representations

- **Entity:** Blue rectangle with white bold text
- **Weak Entity:** Rectangle with double borders (left and right) and yellow fill
- **Subclass:** Purple rectangle with white bold text
- **ISA Relationship:** Triangle connecting superclass to subclasses

2.2 Attribute Representations

- **Attribute:** Circle with green border
- **Primary Key:** Underlined text within the circle

- **Multi-valued Attribute:** Double circle with red boundary
- **Derived Attribute:** Circle with cream-filled background
- **Composite Attribute:** Connected sub-attributes branching from parent
- **Partial Key (Weak Entity):** Underlined with dotted line
- **Relationship Attribute:** Stadium-shaped (rounded rectangle)

2.3 Relationship Representations

- **Relationship:** Orange diamond with white bold text
- **Identifying Relationship:** Orange diamond with dotted border
- **Total Participation:** Thick line (double line in standard notation)
- **Partial Participation:** Normal single line
- **Cardinality:** Indicated using (min, max) notation on relationship lines

3 Design Quality Analysis

We conducted a comprehensive analysis of our ER diagram to identify potential data quality issues and anomalies.

3.1 Redundancy Analysis

Our design exhibits minimal redundancy due to careful normalization principles. The same information is not stored in multiple places within our database. By strategically moving the `Bounty` attribute from the `PIRATE` entity to the `BOUNTY_RECORD` weak entity, we eliminated potential redundancy where bounty values could have been duplicated across both entities. Each bounty value is now stored exactly once with its corresponding issue date and version. Additionally, we utilize derived attributes such as `Age` (computed from `Date_of_Birth`) and `Total_Bounty` (computed from sum of crew member bounties), which are calculated dynamically rather than stored, further preventing any data redundancy.

If redundancy existed in our design, it could cause several problems including data inconsistencies when the same information stored in multiple locations diverges, increased storage requirements, higher maintenance overhead, and difficulties in ensuring data integrity across updates.

3.2 Update Anomaly

Yes, our design contains a potential update anomaly related to ship ownership transfers. If we need to change ship ownership information, we would have to update it in multiple places.

Scenario/Example: When a ship changes ownership from one crew to another (due to crew disbandment or transfer), we need to update the `Owning_Crew_ID` in the `SHIP`

table. If the CREW entity also stores ship-related information like a flagship identifier, we must update multiple places: the ship's owner field and both the old crew's and new crew's flagship records. If these updates are not done together, the database becomes inconsistent—the ship might show it belongs to Crew A, while Crew B's records claim that same ship as their flagship, creating conflicting intelligence reports.

3.3 Insertion Anomaly

Yes, there are situations where we cannot insert data because other related data doesn't exist yet.

Scenario: A significant insertion anomaly exists in our design regarding ship commissioning. We cannot insert a SHIP record into the database without simultaneously assigning a captain, as the CAPTAINS relationship has total participation with cardinality (1,1) from the ship's side. This means if a ship is under construction, newly commissioned, or between captains due to reassignment, we cannot record the ship in the database until a Person_ID is designated as its captain. This constraint forces us to either delay recording valuable intelligence about new vessels or artificially assign a temporary captain placeholder just to satisfy the database constraint. This is problematic because in real-world scenarios, ships may exist in drydock or be awaiting captain assignment, yet we cannot capture this intelligence until the captain relationship is established.

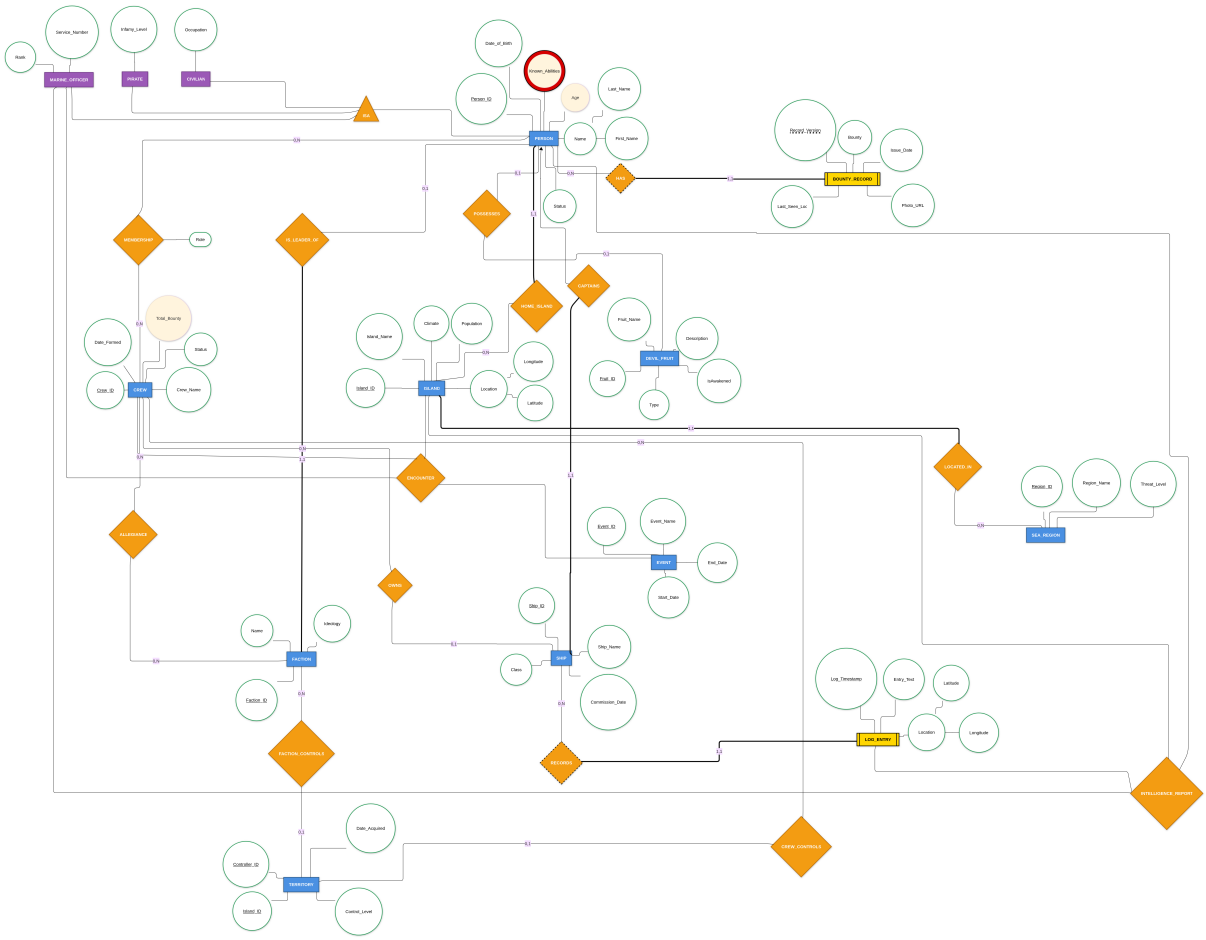
3.4 Deletion Anomaly

Yes, if we delete certain data, we would unintentionally lose other important information.

Example: A critical deletion anomaly exists when deleting a SHIP from the database. If a ship is decommissioned, sunk, or otherwise removed from active service and its record is deleted, all associated LOG_ENTRY records would be cascade deleted due to the identifying relationship between SHIP and the weak entity LOG_ENTRY. These log entries contain valuable historical intelligence—observations of enemy movements, encounter reports, strategic reconnaissance data, and activities recorded throughout the ship's operational lifetime. Deleting the ship would unintentionally cause the permanent loss of this irreplaceable strategic intelligence data, making it impossible to conduct historical analysis of events involving that vessel, track patterns of pirate activity in specific regions, or verify past intelligence reports. This represents a significant loss of institutional knowledge that could be critical for future strategic planning.

4 ER Diagram

The complete Entity-Relationship diagram for the Aegis Intelligence Database is presented below, illustrating all entities, relationships, attributes, and constraints defined in our Phase 1 requirements document with the modifications discussed in Section 1.



Interactive ER Diagram View:
`https://www.mermaidchart.com/d/6d518f6c-2725-48fb-ac00-4944a5a06bf0`